Investigating the Impact of Noise and Outliers on Support Vector Machine Decision Boundaries

MODULE: Machine Learning and Neural Networks
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ABSTRACT

Support Vector Machines (SVMs) are widely used for classification tasks due to their ability to construct optimal decision boundaries by maximising the margin between classes. However, their performance can be significantly affected by data quality, particularly the presence of noise and outliers.

This study investigates how noise and outliers influence SVM behaviour using a synthetic dataset. The model is evaluated under three conditions: clean data, feature noise, and outlier contamination. Additionally, the effect of the regularisation parameter $C$is analysed to understand its role in controlling model robustness.

The results show that SVM performs well on clean data but becomes increasingly unstable when noise is introduced. The most significant distortion occurs with outliers, which strongly influence the decision boundary by becoming support vectors. These findings highlight the sensitivity of SVMs to data quality and emphasise the importance of preprocessing and parameter tuning for reliable performance.

INTRODUCTION

Machine learning models are widely applied in real-world scenarios such as healthcare, finance, and computer vision, where data is often imperfect. These datasets frequently contain noise, measurement errors, and outliers, which can significantly affect model performance.

Support Vector Machines (SVMs), introduced by Cortes and Vapnik (1995), are powerful classification algorithms that construct decision boundaries by maximising the margin between classes. This property allows SVMs to achieve strong generalisation performance under ideal conditions. However, because the decision boundary depends on a subset of data points known as support vectors, SVMs can be sensitive to unusual or extreme observations.

Understanding how SVMs behave under noisy conditions is important for evaluating their robustness. If the model is highly influenced by noise or outliers, its reliability in real-world applications may be limited.

This study investigates how different types of data contamination affect SVM performance by analysing decision boundaries, classification accuracy, and support vector behaviour across multiple scenarios.

BACKGROUND OF SVM

Support Vector Machines are supervised learning algorithms used for classification and regression tasks. The key idea is to find a hyperplane that separates classes while maximising the margin between them.

The margin is determined by the closest data points to the decision boundary, known as **support vectors**. These points define the position of the boundary and are critical to the model's behaviour.

In practical applications, perfect separation is rarely possible. Therefore, SVM introduces a **soft margin**, controlled by the regularisation parameter C. This parameter determines the trade-off between:

- Maximising the margin
- Minimising classification errors

A small value of C allows more misclassification and produces a smoother decision boundary, while a large value of C attempts to classify all points correctly, potentially leading to overfitting. A key limitation of SVM is its sensitivity to noise and outliers. Since support vectors determine the boundary, even a small number of extreme points can significantly influence the model.

DATASET AND EXPERIMENTAL SETUP

The dataset used in this study was generated using the make_moons function from the Scikit-learn library. This dataset contains two interleaving crescent-shaped clusters and is not linearly separable, making it suitable for evaluating non-linear classifiers.

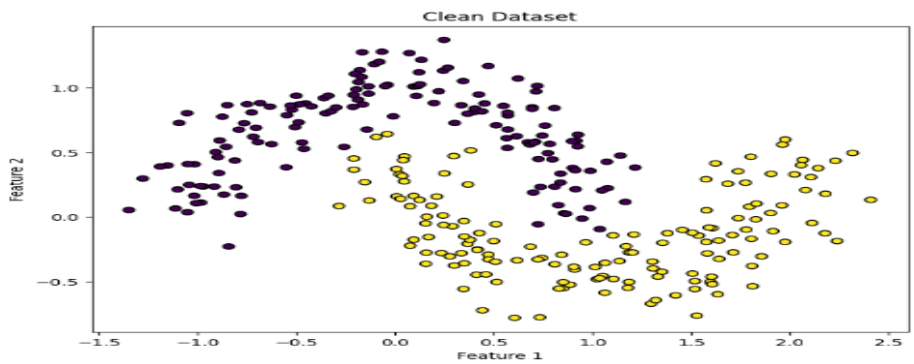
The dataset was chosen because it allows clear visualisation of decision boundaries and makes it easy to observe how the model behaves under different conditions.

The analysis was implemented using Python with the following tools:

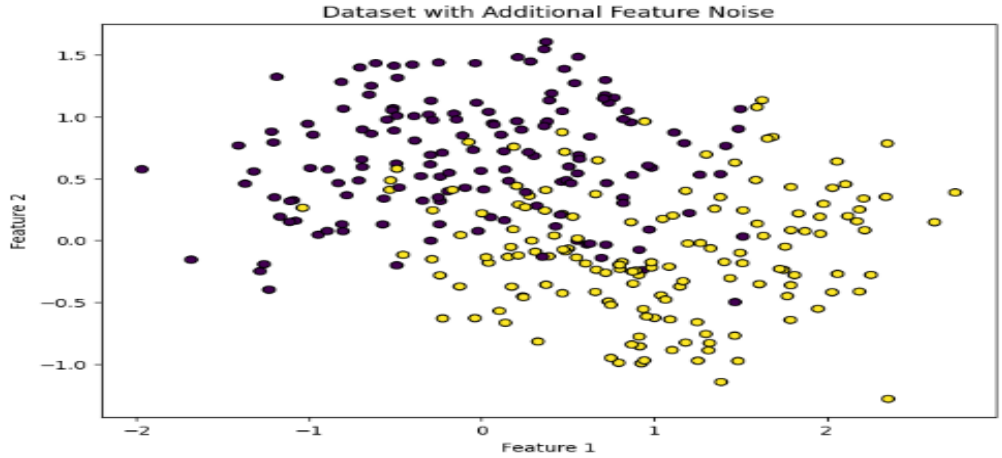
- NumPy for numerical operations
- Matplotlib for visualisation
- Scikit-learn for model implementation

Three experimental scenarios were created:

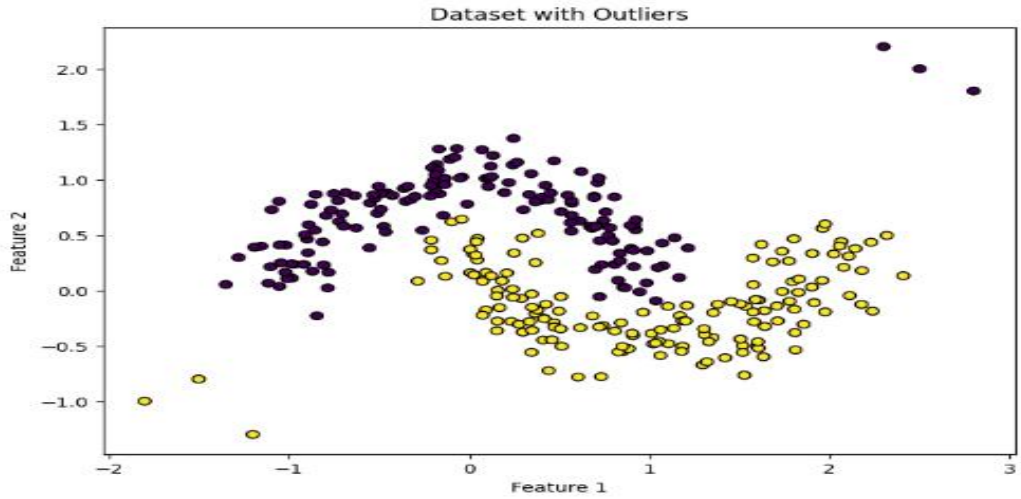
- Clean dataset (baseline)



- Dataset with feature noise



- Dataset with outliers

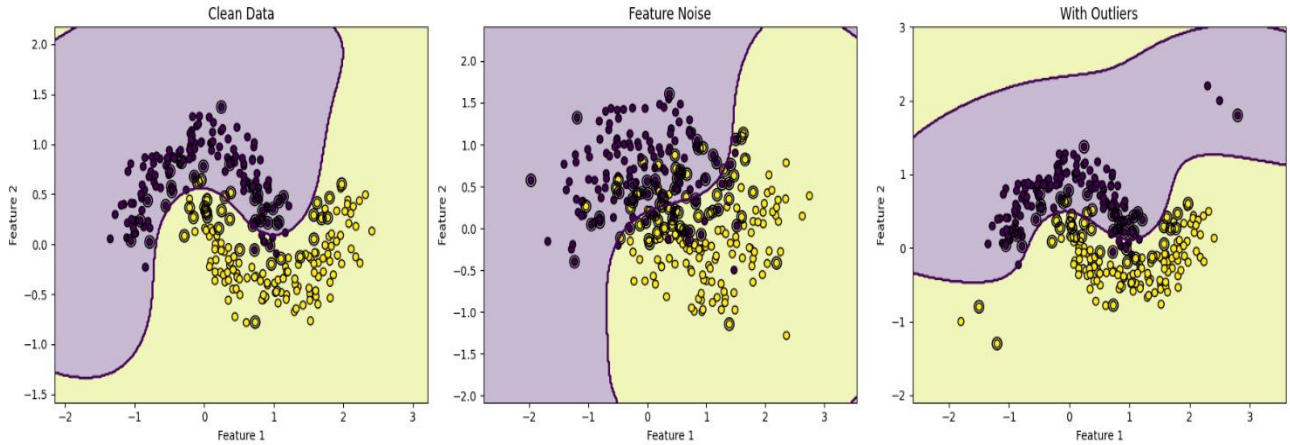


Feature scaling was applied before training, and the dataset was split into training and testing sets.

RESULTS AND VISUALISATION

1. Dataset Distribution

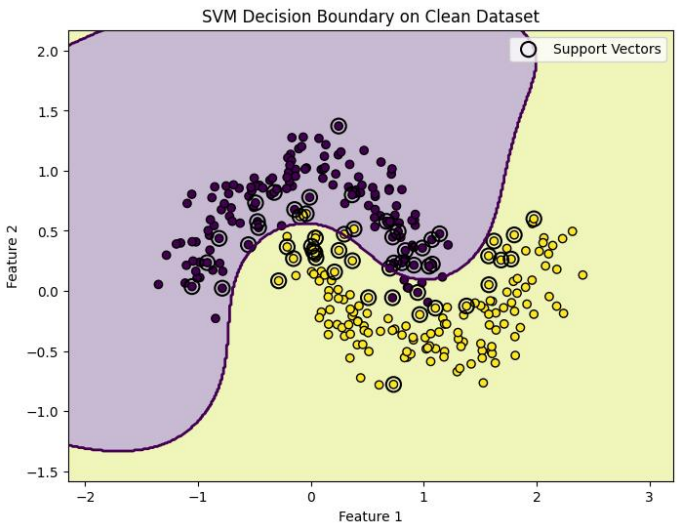
Figure 1: Distribution of the synthetic dataset.



The structure of the dataset shows two curved clusters that cannot be separated by a linear boundary. This makes it suitable for evaluating the performance of non-linear SVM models.

2. Decision Boundary on Clean Data

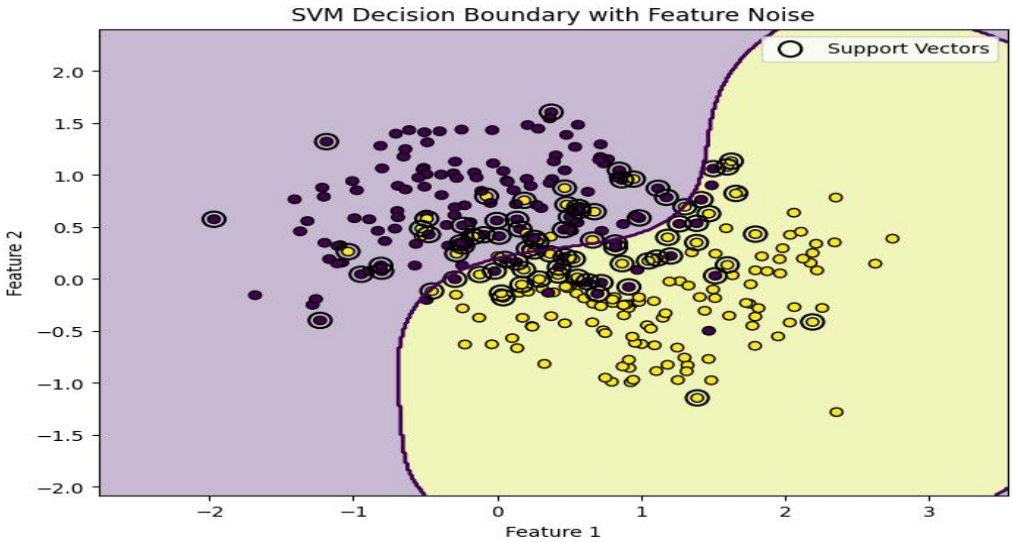
Figure 2: SVM decision boundary on clean dataset.



The model produces a smooth and well-defined boundary that follows the structure of the data. The support vectors lie near the margin, indicating that the model has learned an optimal separation.

3. Effect of Feature Noise

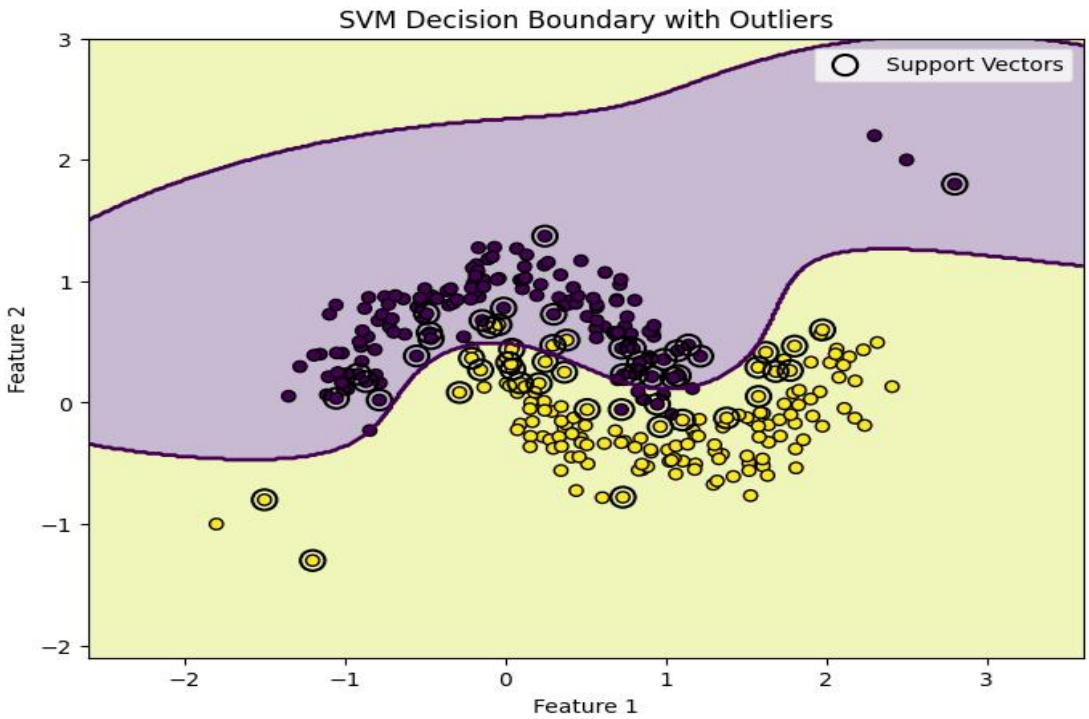
Figure 3: Decision boundary with feature noise.



When noise is introduced, the decision boundary becomes slightly distorted. Although the model still captures the general structure, classification performance decreases due to the increased variability in the data.

4. Effect of Outliers

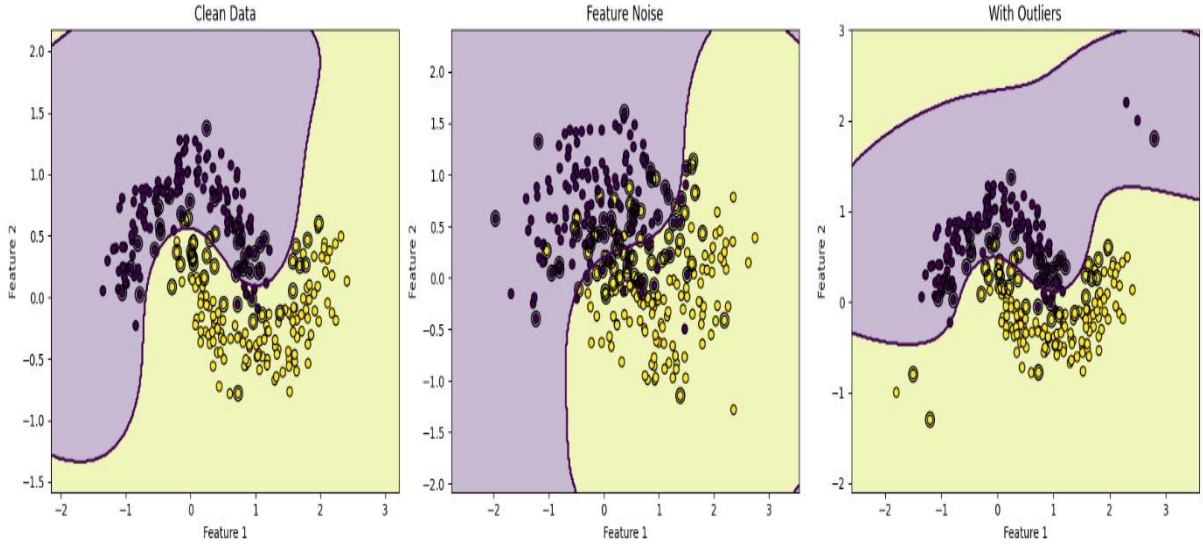
Figure 4: Decision boundary with outliers.



The presence of outliers significantly affects the decision boundary. The boundary shifts towards the outliers, demonstrating that these points strongly influence the model by becoming support vectors.

5. Comparison of Scenarios

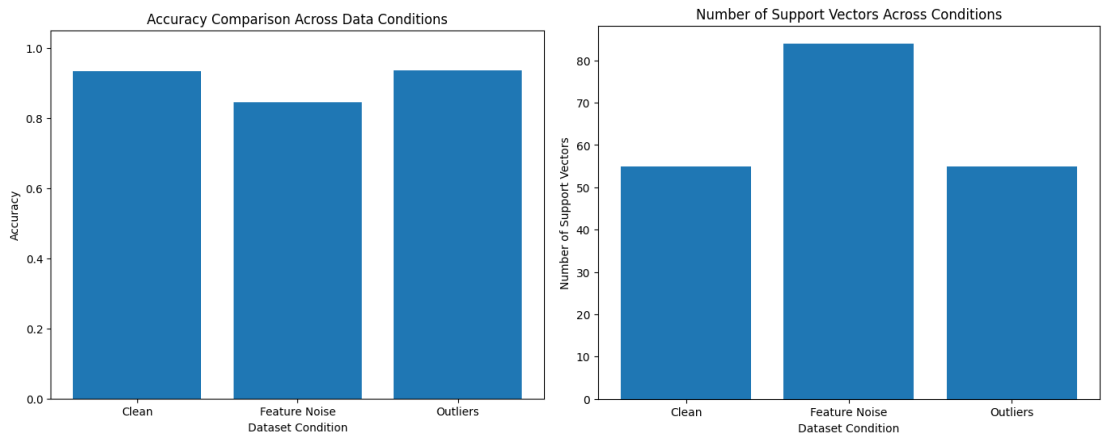
Figure 5: Comparison of clean, noisy, and outlier datasets.



This comparison highlights how data quality impacts the model. The clean dataset produces the most stable boundary, while the outlier dataset shows the greatest distortion.

6. Accuracy Comparison

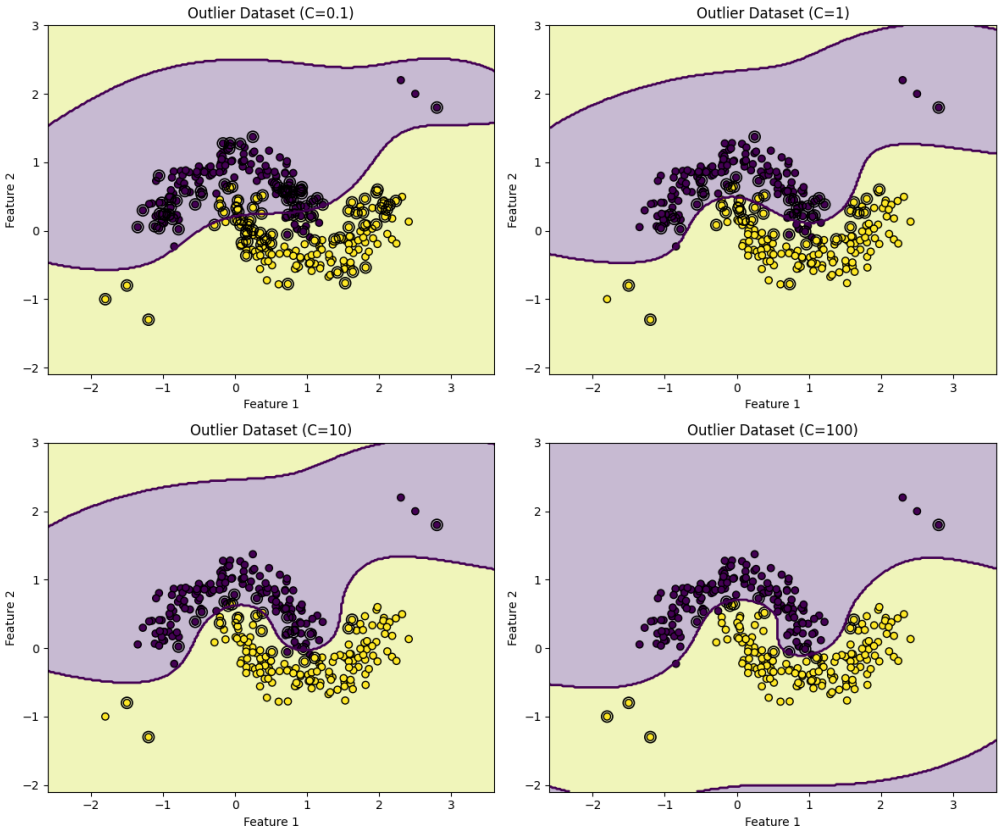
Figure 6: Classification accuracy across different conditions.



The model achieves the highest accuracy on clean data. Accuracy decreases with noise and drops further in the presence of outliers, confirming that data contamination negatively affects performance.

7. Effect of Regularisation Parameter

Figure 7: Decision boundaries for different values of C.

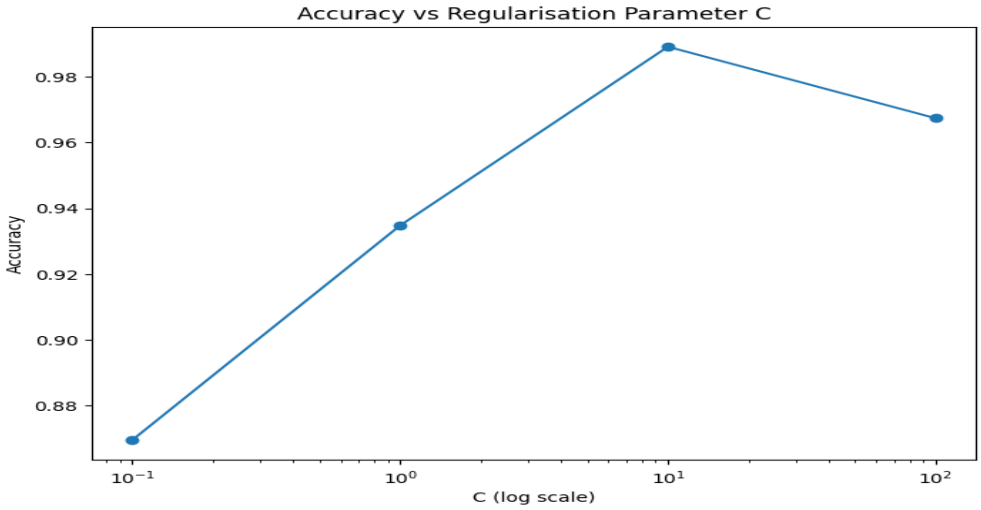


The results show that:

- Small C: smoother and more robust boundary
- Large C: complex boundary sensitive to noise

This demonstrates the importance of hyperparameter tuning in controlling model behaviour.

Figure 8: Accuracy vs Regularisation Parameter C



DISCUSSION

The results demonstrate that SVM performance is strongly affected by data quality.

The model performs best on clean data, where the decision boundary accurately separates the classes. However, when noise is introduced, the boundary becomes less stable and accuracy decreases. This indicates that the model is sensitive to small variations in input features.

The most significant effect is observed with outliers. These extreme data points distort the decision boundary because they often become support vectors. Since support vectors define the model, even a small number of outliers can have a large impact.

The regularisation parameter C plays an important role in controlling this sensitivity. Smaller values of C produce smoother boundaries and improve robustness, while larger values lead to overfitting and increased sensitivity to noise.

Overall, the findings show that SVMs are powerful but require careful handling of data quality and parameter selection.

LIMITATIONS

This study has several limitations. The dataset is synthetic and relatively simple, which may not fully represent real-world data complexity. Only one kernel (RBF) was used, and other kernels may behave differently. Additionally, the evaluation focused primarily on accuracy and did not consider other performance metrics such as precision or recall.

ETHICAL AND REAL-WORLD IMPLICATIONS

The sensitivity of SVM to noise and outliers has important implications for real-world applications. In critical domains such as healthcare or finance, incorrect predictions due to

poor data quality can have serious consequences. Ensuring data reliability and robust model design is essential for safe and ethical AI deployment.

ACCESSIBILITY CONSIDERATIONS

All visualisations were designed with clear labels, readable fonts, and high contrast to ensure accessibility. Figures are described in the text to support understanding.

CONCLUSION

This study investigated how noise and outliers affect Support Vector Machine decision boundaries. The results show that while SVM performs well on clean data, its performance degrades as data quality decreases.

Outliers have the most significant impact because they directly influence the decision boundary through support vectors. The regularisation parameter C plays a key role in controlling this sensitivity, highlighting the importance of hyperparameter tuning.

Overall, the findings emphasise that data preprocessing and careful model configuration

are essential for applying SVM effectively in real-world scenarios.

REFERENCES

- Cortes, C. and Vapnik, V. (1995) *Support-vector networks*.
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